Gaussian-AHP and Machine Learning algorithms to model flood susceptibility for areas with limited inventories

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Abstract - This study developed a novel combinatorial approach for flood susceptibility modeling in urban areas with limited historical flood inventory data. The methodology was based on the application of the Analytic Hierarchy Process-Gaussian (AHP-G) for the generation of training samples, which were subsequently used in machine learning algorithms. The study area was the city of Belém, located in the eastern Brazilian Amazon. A spatial database was constructed using a range of geographic conditioning factors, including elevation, slope, aspect, stream power index, topographic wetness index (TWI), height above the nearest drainage (HAND), distance to drainage channels, and flow accumulation area. The Random Forest (RF), Support Vector Machine (SVM), and Generalized Linear Model (GLM) algorithms were applied to produce flood susceptibility maps. Model validation was conducted using in-situ flood occurrence data collected between 2010 and 2023. The reliability of the models was assessed using various statistical performance metrics, with particular emphasis on the area under the receiver operating characteristic curve (AUC-ROC). The results demonstrated that the Random Forest (RF) algorithm achieved the highest predictive accuracy, with an AUC exceeding 90%. The most influential factors in the modeling process were elevation, HAND, soil characteristics, distance to drainage channels, and precipitation. Overall, the study provides robust and high-quality results that can support decision-making processes regarding future flood risk management and urban planning.

Keywords—Urban, Flood, population, models, planning.

I. Introduction

Floods are among the most frequent and destructive natural hazards worldwide, causing significant environmental, infrastructural, and public health impacts, particularly in urban areas [1]; [2]. Defined as the temporary inundation of normally dry land, often adjacent to natural or artificial drainage systems [3], floods can lead to the spread of waterborne diseases, infrastructure damage, contamination of water sources, and loss of human life. Between 2000 and 2019, over 3,500 flood-related disasters affected more than 2 billion people globally and resulted in approximately US$650 billion in damages (CRED). Projections linked to climate change, land-use transformation, and population growth indicate an increase in both the frequency and severity of flooding events by 2050, with heightened socio-economic losses [4]; [5]. In Brazil, floods remain a critical concern, with more than 3.7 million people affected and over US$3.77 billion in damages reported between 2003 and 2020 (MDR). The city of Belém, capital of Pará state in northern Brazil, exemplifies this vulnerability, with frequent urban flooding driven by factors such as low-lying topography, unplanned urban expansion, insufficient drainage infrastructure, and high rainfall intensity.

The flood risk scenario is further intensified by limited investment in preventive infrastructure, such as drainage systems, and insufficient public education on flood response strategies. Susceptibility, in this context, refers to the degree to which natural or human systems are prone to adverse impacts from external stressors like climate change and extreme weather events, reflecting a system's inherent vulnerability and limited adaptive capacity [6]. Physical-environmental susceptibility models have become essential tools in urban land-use planning and environmental management, enabling the spatial monitoring of areas most affected by hydroclimatic hazards [7]. Predictive susceptibility modeling is increasingly recognized for its role in flood damage mitigation across different spatial and temporal scales [8]. However, traditional rainfall-runoff models, which are predominantly one-dimensional and fail to account for the dynamic and morphologically unstable nature of river systems, often prove inadequate [9]. These methods also require extensive fieldwork, long-term historical data, and substantial financial resources for data acquisition, limiting their practical applicability in many regions [10].

Conversely, machine learning (ML) algorithms have significantly contributed over the last two decades to advancing more precise forecasting systems capable of producing superior results at a reduced financial expense [11]. ML algorithms encompass object-oriented prediction models capable of mathematically capturing the non-linearity of flood processes through historical recurrence data, obviating the necessity for complex physical parameters [12]. The primary advantage of ML algorithms lies in their independence from expert subjectivity, yet their accuracy remains entirely contingent upon the quality of the inventory data.

Flood susceptibility analysis faces significant challenges, with the lack of detailed and georeferenced historical inventories being one of the most critical, particularly for training and validating predictive models based on machine learning algorithms. This limitation is further exacerbated in regions with low orbital sensor monitoring capacity, due to

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persistent cloud cover and the absence of recent records of major flood events. As an alternative, many studies rely on multi-criteria decision-making (MCDM) methods, such as the Analytic Hierarchy Process (AHP) and the Analytic Network Process (ANP), which, although independent of historical inventories, are inherently subject to the subjectivity of expert judgment during pairwise comparisons [13; 14; 15].

To overcome these limitations, the present study proposes an innovative approach by applying the Gaussian-AHP method to minimize subjectivity in generating training samples. This strategy aims to enhance the robustness of flood susceptibility models based on machine learning. The research focuses on training and evaluating the usability of synthetic (phantom) samples for modeling flood susceptibility in the urban area of Belém, using various machine learning algorithms fed with a set of high-resolution, spatially detailed geographic conditioning factors.

II. STUDY AREA

The study area encompasses Belém, the capital of Pará and a significant metropolis in the Brazilian Amazon, with a population exceeding 1.3 million in 2022. It is located in the estuary of the Amazon River and encircled by tributaries such as the Guamá and Pará rivers “Fig. 1”.

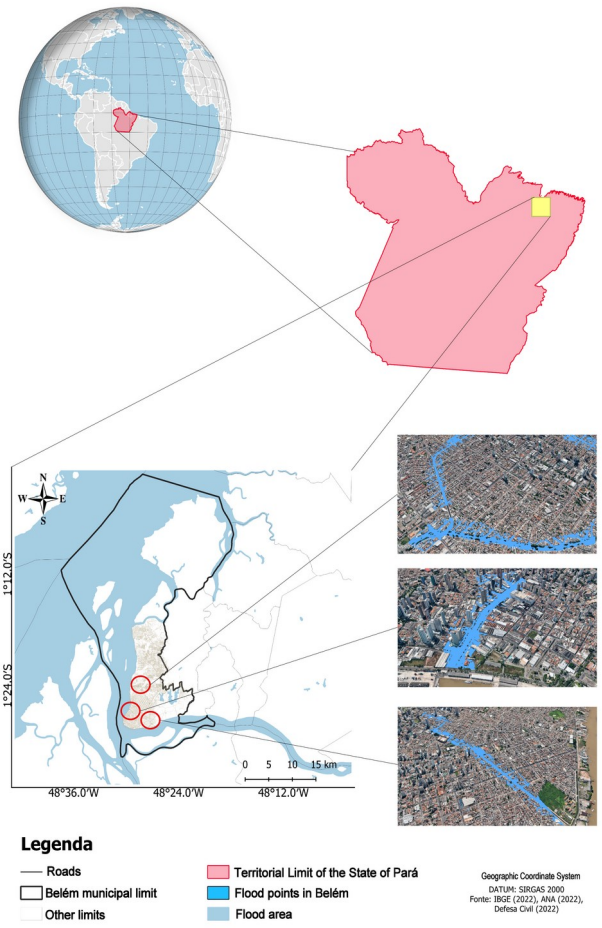


Fig. 1. Location of the research area.

Belém has approximately 1,064 km² and with altitude ranging from 0 to a maximum of 23 meters. It is characterized by a hot and humid equatorial climate, with an average annual temperature of 26°C and annual rainfall exceeding 2,500 mm [16] and its vegetation is predominantly tropical rainforest. Over the past five decades, accelerated urbanization has resulted in population expansion into riverside areas, heightening the risk of flooding and exacerbating socio-environmental issues in the urban area [17].

III. METHODOLOGY

The methodology is summarized in “Fig. 2”. It comprises four main steps: (I) selection of topographic, hydrological, climatic and anthropogenic constraints, (II) generation of training samples derived from AHP-Gaussian and validation samples acquired by survey, (III) implementation of machine learning models and (IV) evaluation of forecast performance.

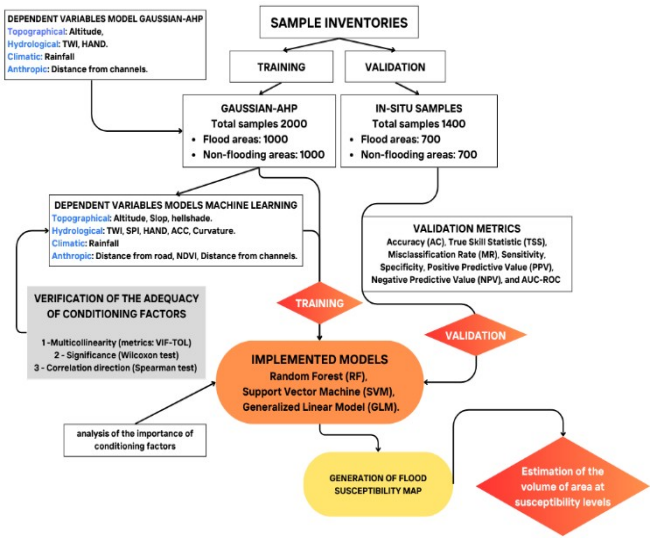


Fig. 2. Flowchart for spatial forecasting of flood susceptibility.

A. *Flood Conditioning Factors*

Flood susceptibility is influenced by a complex interplay of natural, hydrological, morphological, climatic, and anthropogenic factors [3]. This study considered 12 conditioning factors widely cited in the literature [7]; [18]; [19] and relevant to the local context.

Topographical variables included slope, elevation, plane curvature, and aspect, all derived from a high-resolution (2m) Digital Elevation Model (DEM) generated from planialtimetric data provided by the Belém Municipal Works and Planning Department using QGIS 3.22. Slope and aspect influence surface runoff and water infiltration [19]; [20], while elevation shows an inverse relationship with flood susceptibility [21]. Curvature affects flow acceleration and sedimentation patterns [22]**;** [23].

Hydrological and morphological factors included the Topographic Wetness Index (TWI), Stream Power Index (SPI), Height Above Nearest Drainage (HAND), stream accumulation area (ACC), and distance from channels. TWI and SPI were calculated using morphometric parameters from the DEM [24]; [25], while HAND was generated using INPE’s standalone C++ application for hydrological terrain modeling [26]; [27]. These variables reflect water accumulation potential, gravitational drainage capacity, and flow energy [28]**.**

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Rainfall was included as a climatic factor, derived from daily precipitation data interpolated in QGIS using the Inverse Distance Squared method [28]. Anthropogenic factors included distance from roads and the Normalized Difference Vegetation Index (NDVI). Proximity to roads indicates areas of reduced infiltration due to impervious surfaces [29]; [30], while NDVI, obtained from Planet multispectral imagery, reflects vegetation cover and its mitigating role against flood susceptibility [31]; [32]. All spatial layers were processed and standardized using QGIS tools, ensuring consistency across data inputs for subsequent flood susceptibility modeling.

B. Preparation of Training and Validation samples

In the data acquisition process, creating the affected areas inventory is the first step in susceptibility mapping [21]. In this study, a hybrid methodology was developed that integrates the Gaussian-AHP method with machine learning (ML) algorithms for flood susceptibility modeling “Fig. 3”.

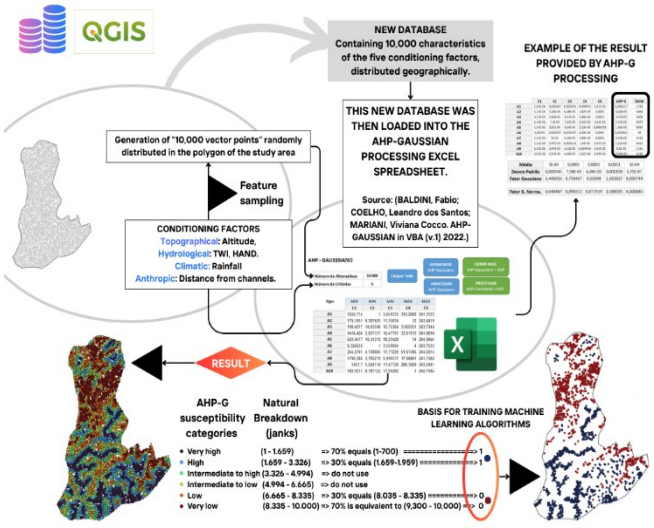


Fig. 3. Flowchart of sampling processing by AHP-G

The traditional Analytic Hierarchy Process (AHP) is a methodological approach to Multi-Criteria Decision-Making (MCDM) that aims to select optimized alternatives based on different criteria and sub-criteria through pairwise comparisons. However, this technique is subject to the inherent subjectivity of the decision-maker or expert, whose perception determines what is considered a priority. The method is structured into three levels: the first defines the overall objective to be achieved, the second comprises the established criteria, and the third consists of the sub-criteria or alternatives to be evaluated [33].

The traditional AHP is typically implemented in seven steps. The first step involves the construction of a pairwise comparison matrix among the criteria, in which the decision-maker uses Saaty’s scale to express how two criteria relate to each other. In the second step, the consistency of the matrix is assessed through the calculation of the consistency index and consistency ratio. The third step defines the priority vector of the criteria, obtained by normalizing the matrix. The fourth step mirrors the first but applies to the alternatives with respect to each criterion. In the fifth step, the consistency of the pairwise comparison matrices of the alternatives for each criterion is verified, again using the same formulas applied in step two. The sixth step establishes the priority vectors of the alternatives. Finally, in the seventh step, the priority vectors of the criteria and the alternatives are multiplied to obtain the global priorities of the alternatives [33]; [34].

In contrast, the Gaussian-AHP is a variation of the traditional method designed to eliminate the subjectivity of pairwise comparisons by calculating the weights of the criteria based on the statistical distribution of the cardinal input data (mean and standard deviation), which is a prerequisite for its application. Like the traditional AHP, this approach also follows seven steps: (i) definition of the decision matrix; (ii) calculation of the mean of the alternatives for each criterion; (iii) calculation of the standard deviation of the criteria based on the samples of the alternatives; (iv) weighting of the decision matrix; (v) normalization of the results; and (vi) derivation of the final ranking of the alternatives [33].

. With this in mind, five conditioning factors suitable for this requirement (precipitation, HAND, TWI, distance to drainage channels, and elevation) were selected to construct a decision matrix derived from 10,000 randomly sampled point locations within the study area.

The Gaussian-AHP was then applied to objectively generate a susceptibility ranking. Subsequently, the results were spatialized in QGIS and filtered using the Natural Breaks (Jenks) classification method, reducing the dataset to 2,000 samples categorized into a binary condition (1,000 representing susceptible areas, labeled as 1, and 1,000 non-susceptible areas, labeled as 0), which served as the training base for the ML models.

In order to validate the models derived from the combination of AHP-Gaussino and ML algorithms, this study used information on areas affected by floods, provided as georeferenced vector data by the Municipal Civil Defense of Belém. The vector data represented water spots mapped during the 2010 and 2023 flood events, which were used to generate an inventory of 700 affected locations. Considering the perimeter outside the flood spots, equal points were randomly selected to represent samples of non-flooded areas. Therefore, 1,400 location points were used in the validation dataset. The data were then categorized into 'flood = 1' and 'non-flood = 0' and incorporated into the validation routine.

C. Adequacy of Conditioning Factors

Initially, a conditioning factor matrix at a spatial resolution of 5 m/pixel and an inventory of vector points identifying flood and non-flood areas were created and projected to the SIRGAS 2000 Datum using UTM Zone 23 South. Both datasets were imported into RStudio (version 4.3.1) for analysis. To ensure the suitability of the conditioning factors for modeling, multicollinearity was assessed using the Variance Inflation Factor (VIF) and Tolerance Level (TOL), where VIF values exceeding 5 and TOL values below 0.1 indicated the need for variable exclusion [34].

Following this, the relationship between conditioning factors and susceptibility classes was evaluated using boxplots and the Wilcoxon rank-sum test for two independent samples, which is appropriate for non-normally distributed data. Significant differences between classes were determined by p-values below 0.05 [35]. Additionally,

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Spearman's rank correlation coefficient was applied to assess monotonic relationships between each conditioning factor and the binary flood susceptibility variable, with interpretation based on the strength and direction of correlations. All statistical analyses were performed within the RStudio environment [19].

D. Machine Larning Algorithms

In this study, flood susceptibility mapping was performed using three widely recognized machine learning (ML) algorithms: Random Forest (RF), Support Vector Machine (SVM), and Generalized Linear Model (GLM). These models were trained with binary-classified samples generated through the AHP-Gaussian method. The RF algorithm, implemented via the 'RandomForest' package in RStudio, is an ensemble-based classifier known for its robustness and accuracy, utilizing 500 trees and six variables per split (mtry) [36]; [37].

The SVM, applied using the 'rminer' package with a radial basis function (RBF) kernel, was parameterized with a constant of 1, gamma of 0.08, and epsilon of 0.1 to handle non-linear separability [38]. The GLM, based on logistic regression with a logit link function from the binomial family, was executed using the 'stats' package [39].

E. Forecast Performance Evaluation

To evaluate the performance of the flood susceptibility models, this study employed both cut-off dependent and cut-off independent metrics. The dependent metrics, calculated from the confusion matrix, included Accuracy (AC), True Skill Statistic (TSS), Sensitivity, Specificity, Positive Predictive Value (PPV), and Negative Predictive Value (NPV). These indicators assess how well the models classify flood and non-flood pixels based on true positives (VP), false positives (FP), true negatives (VN), and false negatives (FN).

The Kappa coefficient was deliberately excluded due to its recognized limitations in model comparison, as discussed by Pôntius and [40] and [41]. Instead, TSS was adopted as a more robust alternative. Additionally, the study utilized the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) as threshold-independent metrics, with AUC values ranging from 0 to 1, where values closer to 1 indicate high model performance and those below 0.5 reflect poor predictive capability [42].

IV. RESULTS AND DISCUSSION

In this study, multicollinearity among the conditioning factors was assessed using Tolerance and Variance Inflation Factor (VIF) values, as presented in Table 1. All VIF values were below 5, with the highest being 3.43 (slope), and Tolerance values remained well above the critical threshold of 0.1, with the lowest recorded at 0.36. These results confirm the absence of multicollinearity among the predictor variables used in the flood susceptibility modeling.

To explore the relationships between the conditioning factors and flooding conditions, Spearman's rank correlation and the Wilcoxon unpaired two-sample test were applied. Spearman's correlation analysis identified a strong negative monotonic relationship between flood occurrence and the variables altitude ($\rho = -0.840$), distance from channels ($\rho = -0.680$), and HAND ($\rho = -0.590$), indicating that higher values of these factors correspond to areas less prone to flooding. This pattern reflects the empirical observation that flood events are concentrated in lower-lying regions and within 100 meters of drainage channels. Similarly, SPI ($\rho = -0.580$), distance from roads ($\rho = -0.428$), and NDVI ($\rho = -0.365$) also showed moderate to strong negative correlations, suggesting reduced flood susceptibility with increasing values "Table.1".

TABLE I. DIAGNOSIS FOR THE CONDITIONING FACTORS

Conditioning Factor	multicolinearidade		Spearman	Wilcoxon
	Tolerance	VIF	direction	P-value
ACC	0,98	1,02	−0.067	0.1114
Aspect	0,99	1,01	−0.187	0.009927
Distance from channels	0,47	2,14	−0.680	< 2.2e-16
Precipitation	0,96	1,04	0.556	0.007328
General curvature	0,94	1,07	−0.058	5,78E-11
Slope	0,36	3,43	−0.078	0.1153
HAND	0,37	2,72	−0.590	< 2.2e-16
Altitude	0,45	2,21	−0.840	0.002009
NDVI	0,54	1,84	−0.365	< 2.2e-16
Distance from roads	0,48	2,06	−0.428	< 2.2e-16
SPI	0,87	1,15	−0.580	< 2.2e-16
TWI	0,41	2,42	0.593	< 2.2e-16

Conversely, TWI ($\rho = 0.593$) and precipitation ($\rho = 0.556$) exhibited positive correlations with flooding, indicating that areas with higher soil moisture potential and precipitation levels are more susceptible to flood events. These findings are consistent with previous studies, such as, which highlight the hydrological significance of these variables in flood modeling [43].

The Wilcoxon test further supported the descriptive capacity of several factors in distinguishing flooded from non-flooded conditions. Statistically significant differences ($p < 0.05$) were found for aspect, distance from channels, precipitation, HAND, altitude, NDVI, distance from roads, SPI, and TWI, indicating their strong potential in characterizing flood-prone environments. In contrast, ACC, general curvature, and slope did not yield significant p-values, suggesting a weaker association with flood dynamics in the study area.

Overall, although all variables remained within acceptable multicollinearity thresholds, only a subset exhibited statistically significant spatial relationships with flooding. This finding is based on the relational behavior between the factors and the sample universe of flood and non-flood conditions, as demonstrated by the Spearman correlation index. By applying a cutoff threshold of 0.5 "Table.1", for both positive and negative correlations, the following variables were identified as relevant: distance to channels, HAND, elevation, TWI, and SPI.

The methodological approach adopted in this study presents inherent limitations and uncertainties related to input data selection, particularly concerning the integration of datasets from varying sources and spatial scales. As highlighted by**,** resampling data to fit a specific resolution can introduce scale-related errors that affect model accuracy [44].

While this study mitigated such issues by utilizing high-resolution spatial data, this choice also entails challenges. Although high-resolution imagery enhances the

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identification of spatial features, it may reduce the generalizability of results across broader or heterogeneous regions due to increased data noise and complexity [45].

This was evident in the analysis of NDVI derived from Planet imagery in Belém, where the fine resolution caused overlap between tree canopies and urban structures, complicating the distinction between permeable and impermeable surfaces. It is worth noting that, although this variable is generally used to assess vegetation health, in this case it served to demonstrate the predominance of urbanization, being considered an indicator of the magnitude of the geospatial anthropization process in the study area.

As a result, some urban areas exhibited flood behavior more closely aligned with vegetated zones, suggesting that they may function as hydrological buffers rather than risk zones. In line with [46], who found model performance unaffected by training-validation sample distribution, and following [44], we suggest that expanding the historical flood occurrence dataset could significantly enhance model training and validation accuracy.

A. Predictive Importance of Flood Conditioning Factors

The predictive analysis of flood conditioning factors revealed "Distance from Channels" as the most influential variable across the RF, SVM, and GLM models, emphasizing the spatial concentration of flood occurrences near macro-drainage systems. This was evidenced by 72.5% of flood samples located within 100 meters and 86.7% within 300 meters of drainage channels.

The HAND variable also showed strong predictive relevance due to its capacity to represent microtopographic variations, with 83.8% of flood events occurring in areas less than 2 meters and 76.2% in areas less than 1 meter above the nearest drainage.

In contrast, slope and flow accumulation consistently exhibited low importance, reflecting the predominantly flat terrain (85.3% under 3 degrees) and low runoff potential (97% of the area), where 99.4% of flood points coincided with minimal flow accumulation values.

B. Flood Susceptibility Mapping

The spatial modeling results revealed differences in the extent and localization of flood-prone areas among the three algorithms.

The SVM model classified the largest portion of the territory as flood-susceptible, identifying 30.3% (60.216 km²) of the study area under susceptibility, including 12.2% (24.264 km²) categorized as very highly susceptible. This extensive delineation reflects the model's sensitivity to subtle terrain variations and hydrological factors.

The GLM model outlined a comparatively smaller flood-susceptible area, totaling 17.0% (34.039 km²); however, it allocated the largest absolute area under the very high susceptibility class, covering 12.9% (25.891 km²), which suggests a more concentrated identification of critical risk zones, “Fig. 4”.

The Random Forest (RF) model presented the most conservative spatial output, mapping 20.1% (39.159 km²) as susceptible, with only 4.0% (7.892 km²) designated as very highly susceptible. Across all models, flood-susceptible zones were strongly concentrated along the city's macro-drainage network and in low-lying floodplain regions, reinforcing the significance of hydromorphological context in urban flood risk assessments.

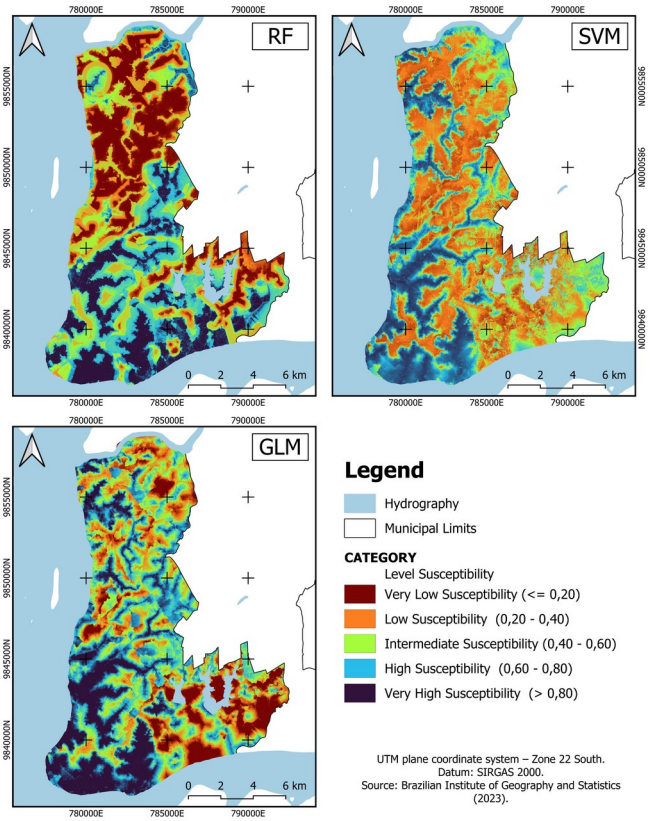


Fig. 4. Flood susceptibility models using: Random Forest, Support Vector machine, Generalized Linear Model.

C. Model Accuracy Validation

The Random Forest (RF) algorithm demonstrated superior performance across all validation metrics compared. Among the tested models, RF achieved the highest positive predictive value (PPV = 0.898) and negative predictive value (NPV = 0.894), indicating a strong ability to correctly classify both flood and non-flood pixels. These results suggest a balanced and accurate distinction between the two classes, reinforcing the robustness of the RF model.

In terms of sensitivity and specificity, the RF model also outperformed its counterparts, recording values of 0.894 and 0.898, respectively. This balance highlights the model's effectiveness in identifying true positive and true negative cases with high precision. In contrast, the SVM model showed a notable discrepancy between specificity (0.884) and sensitivity (0.769), suggesting a tendency to misclassify flood pixels. Similarly, GLM exhibited a specificity of 0.884 but a lower sensitivity of 0.742, indicating limited capability in capturing true positives compared to RF.

When analyzing the cut-off-dependent metrics, the RF model again ranked highest with an accuracy (AC) of 0.896, a true skill statistic (TSS) of 0.864, a misclassification rate (MR) of 0.104, and an F1 Score of 0.896. These results underscore the consistency and reliability of the RF algorithm in predictive classification tasks. The second-best

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performance was observed in the SVM model (AC = 0.827, TSS = 0.686, MR = 0.173, F1 Score = 0.835), followed closely by the GLM model (AC = 0.813, TSS = 0.663, MR = 0.187, F1 Score = 0.824). In terms of overall discriminatory power, as measured by the area under the receiver operating characteristic curve (AUC-ROC), RF once again led with a score of 0.945, confirming its high classification accuracy across thresholds. SVM and GLM followed with AUC-ROC values of 0.882 and 0.868, respectively, indicating reasonably good but comparatively weaker performance.

While all three models provided acceptable results, the RF model consistently demonstrated the highest performance across all key validation metrics. However, it is important to note that conventional machine learning approaches, such as those evaluated in this study, do not inherently account for spatial heterogeneity. As emphasized in prior research [44]**;** [45], incorporating the spatial dimension through geographically weighted or spatially adaptive models can further enhance predictive accuracy in spatial susceptibility assessments.

V. CONCLUSIONS

This study marks a methodological advancement in urban flood susceptibility mapping for data-scarce tropical regions like Belém, eastern Amazon. By integrating a Gaussian Analytical Hierarchy Process for training sample generation with three machine learning algorithms (RF, SVM, and GLM), the research offers a scalable approach to flood risk assessment. The AHP-Gaussian reduced subjectivity by incorporating probabilistic uncertainty, while the Random Forest (RF) model demonstrated the highest predictive performance. The inclusion of HAND as conditioning factor notably enhanced model accuracy by capturing microtopographic controls on flooding.

Despite these advances, there are still limitations, including the need for extreme flood records for reliable long-term reliability. Although AHP-Gaussian reduced bias, it was restricted to working with factors that are represented by cardinal data. Nevertheless, this new approach provides effective susceptibility maps for urban planning and disaster risk management. Future studies should incorporate additional environmental and anthropogenic variables and expand this integrated modeling framework to other Amazonian cities for greater generalizability across climate change scenarios.

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